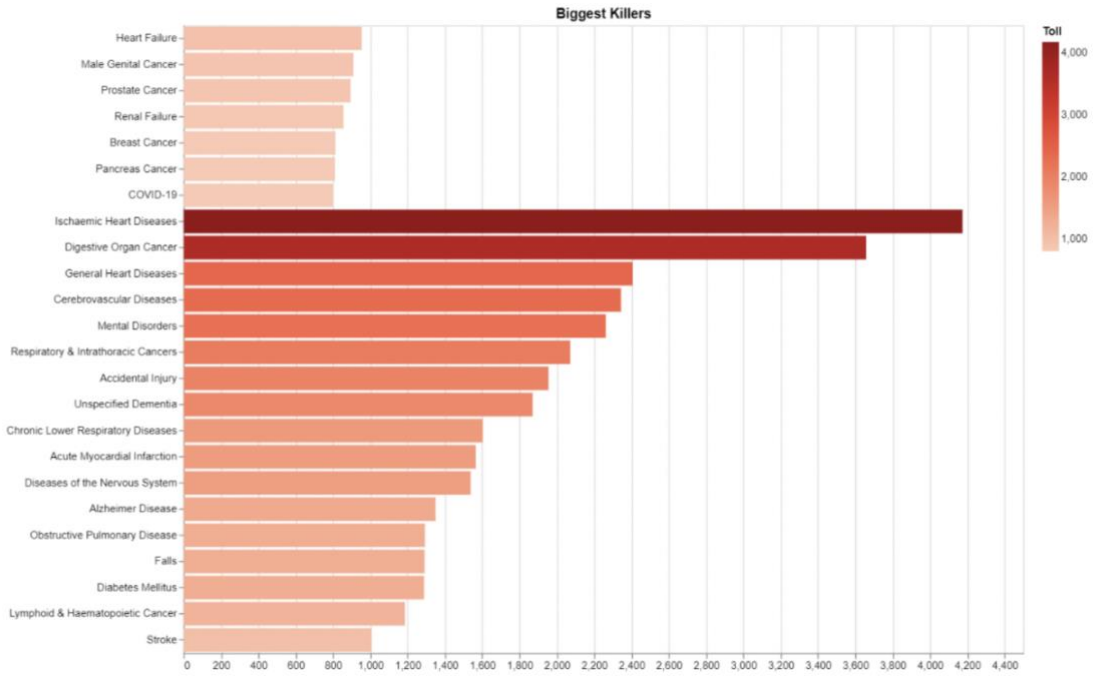
Questions

Q1: Receiving a **warning about an infinite extent for field** “Proportion”, which causes the Vega-Lite editor to not render the map.

Q2: How to **change the column of data used depending on what year is selected**?

Q3: How to **bind different attributes as one field**?

Q1: Receiving a warning about an infinite extent for field "Proportion", which causes the Vega-Lite editor to not render the map.




```
{
  "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
  "title": "Number of Military Personel",
  "width": 1200,
  "height": 900,
  "projection": {"type": "equirectangular"},
  "data": {
    "url": "https://raw.githubusercontent.com/FakeURL.json",
    "format": {"type": "topojson", "feature": "name"}
  },
"params": [
  {
    "name": "Year_selection",
    "value": 2000,
    "bind": {"input": "range", "min": 2000, "max": 2018, "step": 1}
    }
],
  "transform": [
    {
      "lookup": "properties.ISO_A3",
      "from": {
        "data": {
          "url": "https://raw.githubusercontent.com/FakeURL.csv"
        },
        "key": "Country Code",
        "fields": ["Proportion"]
      }
    },
    {"filter": "datum.Year == Year_selection"}
  ],
  "mark": {"type": "geoshape"},
  "encoding": {
    "color": {"field": "Proportion", "type": "quantitative"},
```

```
    "tooltip": [
      {"field": "properties.NAME_EN", "type": "nominal", "title":
"Country"},
      {"field": "Proportion", "type": "quantitative", "title":
"Armed forces personnel (% of total labor force)"}
    ]
  },
  "config": {}
}
```

1. In order to use "datum.Year == Year_selection" in Line 31, you need to include "year" in Line 28. Otherwise, the second data will only have "Proportion" loaded and Vega-Lite cannot find "Year".

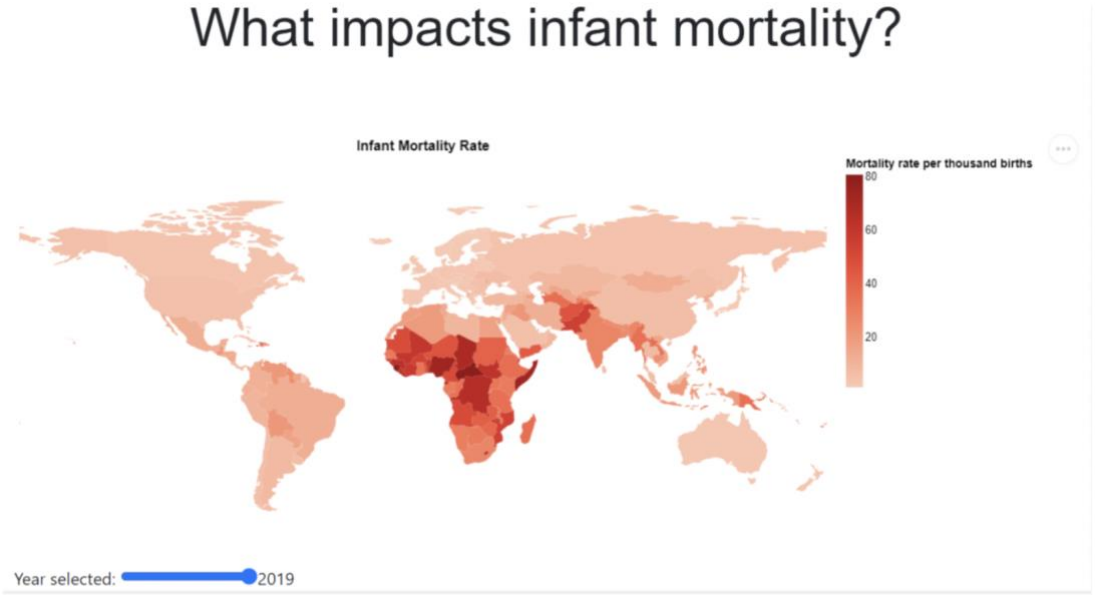
2. If you fix this, you will find that only the year 2018 can be loaded. The problem is that the join in Vega-Lite, for each country in the first dataset, Vega-Lite will only store one row of data from the second table that matches each country. Therefore, only the last year 2018 data will be stored.

To solve this issue, you will need to exchange the position of your Topojson and CSV files, i.e., you need to match a country boundary from the Topojson file for each row of data in the CSV file. The correct code is below.

```
{
  "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
  "title": "Number of Military Personel",
  "width": 800,
  "height": 600,
  "projection": {"type": "equirectangular"},
  "params": [
  {
    "name": "Year_selection",
    "value": 2018,
    "bind": {"input": "range", "min": 2000, "max": 2018, "step": 1}
    }
],
  "data": {
    "url": "https://raw.githubusercontent.com/FakeURL.csv"
  },
  "transform": [
    {
      "lookup": "Country Code",
      "from": {
        "data": {
          "url": "https://raw.githubusercontent.com/FakeURL.json",
          "format": {"type": "topojson", "feature": "name"}
        },
        "key": "properties.ISO_A3"
      },
      "as": "geo"
    },
    {"filter": "datum.Year == Year_selection"}
  ],
  "mark": {"type": "geoshape"},
  "encoding": {
```

```
    "shape": {
        "field": "geo",
        "type": "geojson"
      },
    "color": {"field": "Proportion", "type": "quantitative"},
    "tooltip": [
      {"field": "properties.NAME_EN", "type": "nominal", "title":
"Country"},
      {"field": "Proportion", "type": "quantitative", "title":
"Armed forces personnel (% of total labor force)"}
    ]
  },
  "config": {}
}
```

Q2: How to change the column of data used depending on what year is selected?



Here are two solutions:

1. Long format (CSV file): https://edstem.org/au/courses/6022/discussion/609928

   This will be discussed in the week-10 tutorial.

2. Wide-format (CSV file), see below.

   Pay attention to lines 6 to 18, line 69, and line 75. (Highlighted in yellow)

```
{
  "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
  "title": "Median house price per suburb (2010-2020)",
  "width": "container",
  "height": 500,
  "params": [
    {
      "name": "Year_selection",
      "value": 2018,
```

```
      "bind": {
        "input": "range",
        "min": 2010,
        "max": 2020,
        "step": 1,
        "name": "Year:  "
      }
    }
  ],
  "projection": {
    "type": "equirectangular",
    "center": [145, -37.95],
    "scale": 30000
  },
  "layer": [
    {
      "data": {
        "url":
"https://raw.githubusercontent.com/KaneSec/vega_lite/main/6_advanced
_examples/data/VIC_LOCALITY_POLYGON_SHP.json",
        "format": {"type": "topojson", "feature":
"VIC_LOCALITY_POLYGON_SHP"}
      },
      "transform": [
        {
          "calculate": "'Data is not available in ' +
datum.properties.NAME",
          "as": "note"
        }
      ],
      "mark": {
        "type": "geoshape",
        "fill": "#ddd",
        "stroke": "white",
        "strokeWidth": 1
      },
      "encoding": {"tooltip": {"field": "note"}}
    },
    {
      "data": {
        "url":
"https://raw.githubusercontent.com/KaneSec/vega_lite/main/6_advanced
_examples/data/house_price_by_suburb_wide_format.csv"
      },
      "transform": [
        {
          "lookup": "locality",
          "from": {
            "data": {
              "url":
"https://raw.githubusercontent.com/KaneSec/vega_lite/main/6_advanced
_examples/data/VIC_LOCALITY_POLYGON_SHP.json",
              "format": {
                "type": "topojson",
                "feature": "VIC_LOCALITY_POLYGON_SHP"
              }
            },
            "key": "properties.NAME"
```

```
          },
          "as": "geo"
        },
        {"calculate": "datum[Year_selection]", "as": "year_price"}
      ],
      "mark": {"type": "geoshape", "stroke": "#fff", "strokeWidth":
0.5},
      "encoding": {
        "shape": {"field": "geo", "type": "geojson"},
        "color": {
          "field": "year_price",
          "type": "quantitative",
          "scale": {
            "domain": [
              400000,
              1800000
            ],
            "scheme": "reds"
          },
          "legend": {"format": ".2s"}
        },
        "tooltip": [
          {"field": "locality", "type": "nominal", "title":
"Suburb"},
          {
            "field": "price",
            "type": "quantitative",
            "title": "Median Price",
            "format": ","
          },
          {"field": "year", "type": "quantitative", "title": "Year"}
        ]
      }
    }
  ],
  "config": {}
}
```

Q3: How to bind different attributes as one field?

```
{
    "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
    "data": {
        "url": "https://raw.githubusercontent.com/1099238610/global-
map-vis/main/data/modern-renewable-energy-consumption.csv"
    },
    "params": [
        {
            "name": "year_selection",
            "value": "2018",
            "bind": {
                "input": "range",
                "max": 2018,
                "min": 1985,
                "step": 1,
```

```
                "name": "Year selection"
            }
        },
        {
            "name": "power_type",
            "value": "Hydropower",
            "bind": {
                "input": "select",
                "options": [
                    "Hydropower",
                    "Solar",
                    "Wind",
                    "Other"
                ],
                "name": "Power type"
            }
        }
    ],
    "vconcat": [
        {
            "width": 1000,
            "height": 400,
            "title": "Global clean energy power amount (1985 -
2018)",
            "projection": {
                "type": "equirectangular"
            },
            "transform": [
                {
                    "lookup": "Code",
                    "from": {
                        "data": {
                            "url":
"https://raw.githubusercontent.com/1099238610/global-map-
vis/main/json/global.topo.json",
                            "format": {
                                "type": "topojson",
                                "feature": "custom.geo"
                            }
                        },
                        "key": "properties.wb_a3"
                    },
                    "as": "geo"
                },
                {
                    "calculate": "datum[power_type]",
                    "as": "p_type"
                },
                {
                    "filter": "datum.Year == year_selection"
                }
            ],
            "mark": {
                "type": "geoshape"
            },
            "encoding": {
                "shape": {
                    "field": "geo",
```

```
                    "type": "geojson"
                },
                "color": {
                    "field": "p_type",
                    "type": "quantitative",
                    "scale": {
                        "type": "threshold",
                        "domain": [
                            0,
                            10,
                            50,
                            100,
                            500
                        ],
                        "range": [
                            "#FADBD8",
                            "#F1948A",
                            "#EC7063",
                            "#CB4335",
                            "#B03A2E",
                            "#78281F"
                        ]
                    }
                },
                "tooltip": [
                    {
                        "field": "Entity",
                        "type": "nominal",
                        "title": "Country"
                    },
                    {
                        "field": "p_type",
                        "type": "quantitative",
                        "title": "datum.power_type"
                    }
                ]
            }
        },
        {
            "hconcat": [
        {
            "width": 750,
            "height": 200,
            "mark": "line",
            "title": "Global clean energy power amount - top 5
country",
            "params": [
                {
                    "name": "time_select",
                    "select": {
                        "type": "interval",
                        "encodings": [
                            "x"
                        ]
                    }
                }
            ],
            "transform": [
```

```
                {
                    "filter": "datum['Entity'] == 'China' ||
datum['Entity'] == 'Russia'||datum['Entity'] == 'Canada' ||
datum['Entity'] == 'Brazil' ||  datum['Entity'] == 'United States'"
                },
                {
                    "filter": "datum.Year >= 1985"
                },
                {
                    "calculate": "datum[power_type]",
                    "as": "p_type"
                }
            ],
            "encoding": {
                "x": {
                    "field": "Year",
                    "axis": {
                        "title": "Year"
                    }
                },
                "y": {
                    "field": "p_type",
                    "type": "quantitative",
                    "title": "Power(TW/h)"
                },
                "color": {
                    "field": "Entity",
                    "title": "Country",
                    "type": "nominal"
                }
            }
        },
        {
            "mark": {
                "type": "arc",
                "innerRadius": 60
            },
            "transform": [
                {
                    "filter": "datum['Entity'] == 'China' ||
datum['Entity'] == 'Russia'  || datum['Entity'] == 'Canada' ||
datum['Entity'] == 'Brazil' ||  datum['Entity'] == 'United States'"
                },
                {
                    "filter": "datum.Year == year_selection"
                },
                {
                    "calculate": "datum[power_type]",
                    "as": "p_type"
                }
            ],
            "encoding": {
                "theta": {
                    "field": "p_type",
                    "type": "quantitative"
                },
                "color": {
                    "field": "Entity",
```

```
                    "type": "nominal",
                    "title": "Country"
                }
            }
        }
            ]
        },
        {
            "mark": "bar",
            "transform": [
                {
                    "filter": "datum['Entity'] == 'China' ||
datum['Entity'] == 'Russia'  || datum['Entity'] == 'Canada' ||
datum['Entity'] == 'Brazil' ||  datum['Entity'] == 'United States'"
                },
                {
                    "filter": "datum.Year == year_selection"
                },
                {
                    "calculate": "datum[power_type]",
                    "as": "p_type"
                }
            ],
            "encoding": {
              "column": {"field": "year"},
              "x": {"field": "p_type", "type": "quantitative"},
              "y": {"field": "Entity", "type": "nominal"},
              "color": {"field": "power_type", "type": "nominal"}
        }
      }
    ]
}
```

Check the second line chart example here: https://vega.github.io/vega-lite/docs/repeat.html

Here is a stacked bar chart.

```
{
    "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
    "data": {
        "url": "https://raw.githubusercontent.com/1099238610/global-
map-vis/main/data/modern-renewable-energy-consumption.csv"
    },
    "repeat": {
        "layer": ["Wind", "Solar"]
    },
    "spec": {
        "mark": "bar",
        "transform": [{
                "filter": "datum['Entity'] == 'China' ||
datum['Entity'] == 'Russia'  || datum['Entity'] == 'Canada' ||
datum['Entity'] == 'Brazil' ||  datum['Entity'] == 'United States'"
            },
            {
                "filter": "datum.Year == 2000"
            },
```

```
            {
                "calculate": "datum['Solar']",
                "as": "p_type"
            }
        ],
        "encoding": {
            "x": { "field": { "repeat": "layer" }, "type":
"quantitative" },
            "y": { "field": "Entity", "type": "nominal" },
            "color": { "datum": { "repeat": "layer" }, "type":
"nominal" }
        }
    }
}
```

You can put both the repeat and spec parts in your final layer.